

FL Clients

Client 1

(Honest)

$$g_i \sim \mathcal{N}(\nabla F_i, \sigma^2 I)$$

Client 2

(Honest)

$$g_i \sim \mathcal{N}(\nabla F_i, \sigma^2 I)$$

Client 3

(Honest)

$$g_i \sim \mathcal{N}(\nabla F_i, \sigma^2 I)$$

Client 4

(Byzantine)

$$\tilde{g}_i$$

n clients, $f < n/2$

$$\mathbb{E}[\xi_j^2] \leq \sigma^2$$

(1) FD Sketch

$$G \in \mathbb{R}^{n \times d}$$



Frequent Directions

$$\tilde{G} = \text{Sketch}(G, k)$$

$$\tilde{G} \in \mathbb{R}^{k \times d}, k \ll n$$

Full $O(d^2)$ 1428GB

Sketch $O(k^2)$ 140.9GB

↓ 1 × 2193 mem

$$\hat{C} = \frac{1}{n} \tilde{G} \tilde{G}^T$$

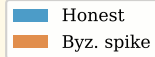
Gram matrix

(2) Eigenspectrum

Eigendecomposition

$$\lambda_1 \geq \dots \geq \lambda_n$$

λ_+



λ

$\hat{\sigma}^2, \hat{\gamma}$ from bulk

$$\lambda_+ = \hat{\sigma}^2(1 + \sqrt{\hat{\gamma}})^2$$

MP upper edge

MP law holds for honest

gradients (Non-IID OK)

(3) KS Test

KS Detection

KS goodness-of-fit:

$$D_{KS} = \sup_{\lambda} |F_n - F_{MP}|$$

Reject if $D_{KS} > \tau_{KS}$

Tail anomaly test:

$$\mathcal{A} = \{i : \lambda_i > \lambda_+ + \tau \hat{\sigma}^2\}$$

Detect if $|\mathcal{A}| > 0$

$$P[\text{det}] \geq 1 - e^{-k/\log^2 k}$$

1-round completeness

Phase trans.: $\sigma^2 f^2 = 0.25$

< 0.25: detectable; ≥ 0.25 : impossible

Sketching error:

$$|\lambda_i(C) - \lambda_i(\tilde{C})| \leq O(k^{-1/2})$$

$k \geq \Omega(\log d)$ sufficient

(4) Filter+Agg

Filter & Aggregate

$$\hat{\mathcal{B}} = \{i : g_i \notin \text{MP bulk}\}$$

$\mathcal{H} = [n] \setminus \hat{\mathcal{B}}$
[OK] C1 incl.

[OK] C2 incl.

[OK] C3 incl.

[X] C4 excl.

Honest-only mean:

$$\hat{g} = \frac{1}{|\mathcal{H}|} \sum_{i \in \mathcal{H}} g_i$$

Convergence:

$$\min_t \mathbb{E}[\|\nabla F\|^2] \leq O\left(\frac{\sigma f}{\sqrt{T}} + \frac{f^2}{T}\right)$$

Minimax optimal

$$T^* = O(\sigma f / \epsilon^2 + f^2 / \epsilon)$$

Blockchain

Self-Stab. Shared Mem.

Round $t-1$: h_{t-1}

Round t : h_t

Round $t+1$: pending

BFT Consensus

$f < n/2$ Byzantine

- * Safety
- * Liveness
- * Closure

Write-once immutable:

No Byzantine rollback.

Polygon / Hardhat

0.15 MATIC / round

Global Model

$$w^{t+1}$$

$$w^{t+1} = w^t - \eta \hat{g}$$

$$\eta = O(1/\sqrt{T})$$

Self-Stabilization:

Any $w^0 \in \mathbb{R}^d$

converges in T^* rounds

Phase transition:

$$\sigma^2 f^2 \geq 0.25$$

impossible (FLP analogy)

Experiment:

78.4% vs 63.4% baseline

Spectral Self-Stabilizing Loop 2014 $T^* = O(\sigma f / \epsilon^2 + f^2 / \epsilon)$ from any w^0